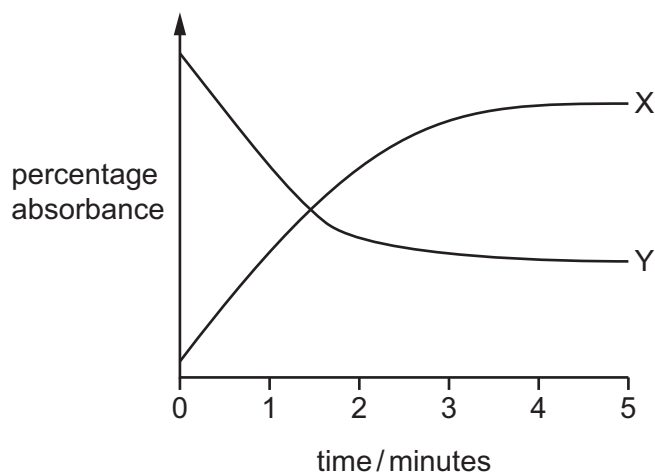


- 13** A scientist investigated the progress of two enzyme-catalysed reactions in separate test-tubes, X and Y. Both reactions result in colour changes that can be detected using colorimetry.

0.5 cm³ samples were taken from each test-tube at the start of the investigation and at regular intervals for the next 5 minutes. Copper ions were added to each sample as soon as the sample was collected to inactivate the enzymes and stop the reactions from progressing further.

The absorbance of each sample was measured using a colorimeter.

The graph shows the results of this investigation.



Which statement is consistent with the results shown in the graph?

- A** The substrate in test-tube X has a higher absorbance than the product.
- B** The product in test-tube Y has a lower absorbance than the substrate.
- C** The rate of the reaction in test-tube X increased with time.
- D** The rate of the reaction in test-tube Y increased with time.